

Integration by Parts

1. The formula for Integration by Parts:

$$\int f(x) g'(x) dx = f(x) g(x) - \int g(x) f'(x) dx \quad \text{OR} \quad \int u dv = u v - \int v du$$

Example 1: Evaluate each of the following integrals:

a) $\int x \cos x dx$

c) $\int x^2 \sin 2x dx$

e) $\int x \sin^2(x) \cos(x) dx$

g) $\int x \sec^2(x) \tan(x) dx$

i) $\int (x+3) \cot^2(2x) dx$

k) $\int (3x+1) (\cos(x) + \sin(x))^2 dx$

b) $\int x \sin x \cos x dx$

d) $\int x^3 \sin \pi x dx$

f) $\int x \sec^2(x) dx$

h) $\int (x+3) \cot^2(2x) dx$

j) $\int (2x^2 - x + 1) e^{-x/2} dx$

l) $\int x^4 e^{2x} dx$

Example 2: Evaluate each of the following integrals:

m) $\int x^3 e^{x^2} dx$

o) $\int \cos(\sqrt[3]{x}) dx$

n) $\int e^{\sqrt{x}} dx$

p) $\int_1^e \sin(\ln x) dx$

Example 3: Evaluate each of the following integrals:

q) $\int x^2 \ln(\sqrt{x}) dx$

r) $\int \ln(x) dx$

Example 4: Evaluate each of the following integrals:

s) $\int e^x \cos(x) dx$

t) $\int e^{2x} \sin(4x) dx$